

Lecture 9: Types of Biases in Data-driven Technology

HUM 4441: Engineering Ethics for Computer Science Engineers

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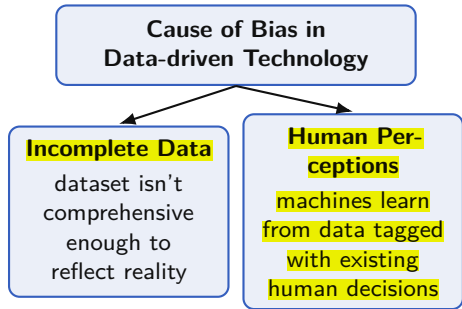
What Is Bias?

Bias

Prejudice in favor of or against one thing, person, or group compared with another. This preference influences understanding and outcomes in a way that **prevents neutrality or objectivity**.

Bias in Machine Learning

Occurs when the predictions or decisions produced by a model are **unfair** - a symptom of the data or the way the problem was framed, not a flaw in the math itself.

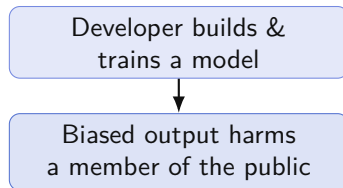


The Software Engineer's Duty

Recall Lecture 5: engineers hold a **professional duty to protect the public**. When the “product” is an algorithm, that duty extends to the people affected by its predictions.

ACM Code of Ethics

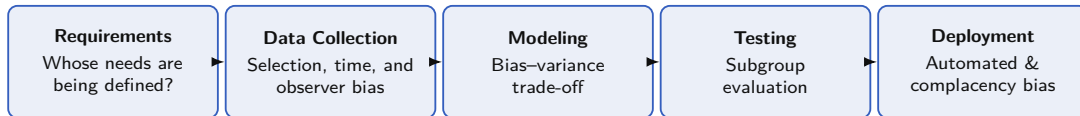
- **1.4** Be fair and take action not to discriminate
- **2.5** Give comprehensive evaluations of systems and their impacts, including possible risks
- **3.1** Articulate social responsibilities and encourage full participation



This is the harm ACM Articles 1.4 and 2.5 exist to prevent.

Bias Across the Software Development Lifecycle

Each stage of building a data-driven system can introduce its own bias, from how the requirements are framed to how the deployed system is monitored, which is why **every stage** needs its own check.



Sociological Bias: Society-Level Prejudice

1 Racism

Belief that one's own race or culture is superior (ethnocentrism).

"British drive on the wrong side of the road."

In practice: racial profiling in predictive-policing tools.

2 Sexism

Belief that one gender is superior.

"Men are stronger and do all the work."

In practice: a hiring model trained on male resumes penalizes women.

3 Ageism

Stereotyping based on age.

"Older people can't use new tech."

In practice: interfaces and health-tech that exclude older users.

4 Religious Prejudice

Irrational fear or hatred of a religion.

"Followers of religion X are dangerous."

In practice: discriminatory content moderation and surveillance systems.

Lecture 9, Types of Bias

Implicit Bias: The Unconscious Kind

Definition

Negative attitudes or stereotypes towards a person or group that **we aren't even consciously aware of**.

- **Not accessible through introspection** - revealed only through implicit association tests
- **May oppose our conscious beliefs** - a believer in equality can still hold implicit bias
- **Pervasive** - everyone possesses them to some degree
- **Caused by exposure (or lack of it)** - stereotypes absorbed from friends, family, or limited knowledge of a group

Affinity Bias

An implicit bias: an unconscious pull toward people who resemble us.

A hiring panel that keeps rating same-background candidates as a better “culture fit” is quietly passing over equally qualified candidates from different backgrounds - not on merit, but on resemblance.

Cognitive Bias: Mental Shortcuts

1 Anchoring

Over-reliance on the **first** piece of information offered.

Doctors diagnosing based on first impressions.

2 False Consensus

The tendency to **over-estimate** how normal our own values and ideas are.

3 Optimism Bias

Believing our own chances of experiencing something negative are **lower** than those around us.

4 Attribution Error

Explaining others' behavior by **character**, not circumstance.

"I was late because of traffic."



"It's because you are lazy."

4 Problems That Biases Help Us Address

Cognitive biases are mental shortcuts the brain evolved to handle a flood of information no one could process in full.

1 Too Much Information

There is too much information in the world - we filter almost all of it out.

2 Not Enough Meaning

We see only a tiny part of the world but must make sense of it to survive.

3 What to Remember?

We can only keep information that is likely to prove useful in the future.

4 Need to Act Fast

We're constrained by time and information, and cannot let that paralyze us.

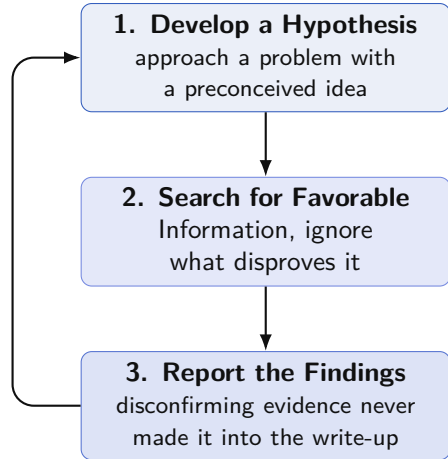
The same shortcuts that help someone get through daily life can just as easily steer a **requirements meeting** or a **data-labeling decision** without anyone noticing.

Confirmation Bias

Definition

The tendency to search for, interpret, favor, and recall information that **confirms** one's pre-existing beliefs.

In data science, this can produce models that confirm our own assumptions rather than discover the true patterns - e.g., collecting data that validates a feature you already believe is important.



Complacency Bias & Automated Bias

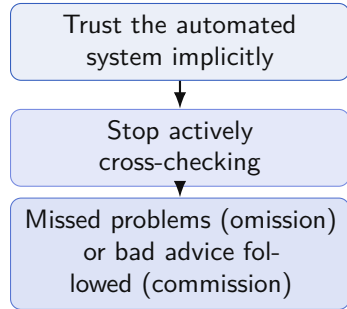
Complacency Bias

Over time, trusting an automated system leads to **reduced vigilance** - the human quietly stops actively cross-checking the system's work at all.

Automated Bias

Faced with a specific recommendation, people **favor the automated system's answer over contradictory evidence** - even their own senses.

This produces two failure modes: an **omission error** (missing a problem the system didn't flag) and a **commission error** (acting on a wrong recommendation the system did give).



Example: GPS routing - drivers follow a wrong turn (commission) even when the road ahead clearly disagrees.

Data Collection Bias

1 Selection

Not randomized; data collection isn't representative.

1936: a poll of car & phone owners predicted the wrong president.

2 Time Interval

A single time period is selected instead of a wide range.

A retailer treats its best holiday month as the year-round average.

3 Observer

The person recording or judging the data unconsciously shapes what gets measured.

The 'Clever Hans' horse: trainers unconsciously cued the "correct" answer.

4 Misclassification

Data is incorrectly classified or labeled.

Mislabeled scans teach a diagnostic model the wrong pattern.

Lecture 9, Types of Bias

1 Outliers

Extreme points get dropped, or kept, for the wrong reasons - skewing the model either way.

*25, 29, 32,
110, 33, 27, 28*

2 Survivorship

Conclusions are drawn only from cases that made it into the dataset; the ones that failed or dropped out are invisible.

WWII bombers: armor went where survivors were hit, not where the losses were hit.

3 Omitted Variable

Leaving out a variable that truly matters lets the model wrongly credit its effect to whatever remaining variable correlates with it.

A pay model without experience makes gender look like the driver of salary.

4 Confounding

Two variables move together only because a hidden third variable drives both - not because either causes the other.

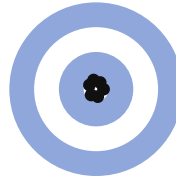
Drowning deaths and ice cream sales both rise with summer heat.

Modeling Bias: The Bias-Variance Trade-off

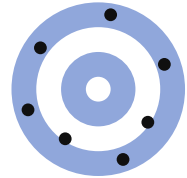
- **Bias:** how well a model matches a training set. A model with **low bias** aligns closely to the training set.
- **Variance:** how much a model changes when you use different portions of your dataset during training. A model with **high variance** is very flexible.
- **Underfit models:** high bias, low variance.
- **Overfit models:** low bias, high variance.

The Goal

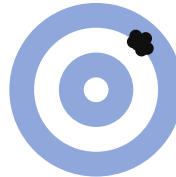
We want **both low bias and low variance** for a well-trained model.



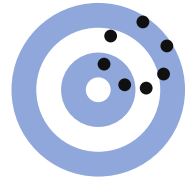
Low Bias
Low Variance



Low Bias
High Variance



High Bias
Low Variance



High Bias
High Variance

Fairness Metrics: Can We Even Define “Fair”?

Once bias is found, engineers still must choose **how to measure fairness** - and different definitions can conflict with each other mathematically.

Demographic Parity

Positive outcome rates are equal across groups, regardless of actual qualification.

Equalized Odds

Error rates (false positive / false negative) are equal across groups.

The Impossibility Result

Except in special cases, a model **cannot simultaneously satisfy** both demographic parity and equalized odds when base rates differ between groups.

Engineering implication: choosing a fairness metric is an **ethical decision** as much as a technical one, so it belongs in the design documentation, justified like any other requirement rather than buried as an unstated default.

An Engineer's Responsibility: Mitigating Bias

- **Audit the data** - check who is, and isn't, represented before training; document dataset limits, as with any test report.
- **Test across subgroups** - evaluate accuracy separately for race, gender, and age, not just one overall average.
- **Diversify the team** - diverse teams are less prone to affinity bias and more likely to catch blind spots early.
- **Document & disclose** - publish known limitations and risks even when they complicate the marketing story; this is a **conscientious** engineering call.
- **Don't automate blindly** - keep a human in the loop; guard against complacency and automated bias in deployment.
- **Speak up** - if management ignores a known bias risk, recall Lecture 5: this may be grounds for **whistle-blowing**.

Case Study: The COMPAS Recidivism Algorithm

The System

COMPAS scored defendants' risk of reoffending, informing U.S. judges' bail and sentencing decisions.

The Investigation

A 2016 ProPublica analysis found Black defendants were far more likely to be **falsely flagged as high-risk**, and white defendants more likely to be **falsely flagged as low-risk**.

The Hard Ethical Question

The vendor countered that the tool was **equally accurate overall** for both groups - illustrating the fairness-metric conflict from two slides ago: equal accuracy does not mean equal error types.

Northpointe, the vendor, never disclosed the algorithm's exact weighting, so neither side's claim could be independently verified - a trade secret shaping a person's liberty, with no public audit trail.

Summary

Human Origins

- Sociological bias
- Implicit / affinity bias

Individual Thinking

- Anchoring, false consensus, optimism, attribution error
- Confirmation bias

The Data Pipeline

- Selection, time interval, observer, misclassification
- Outliers, uncorrelated data

The Model & The Engineer

- Complacency & automated bias
- Bias-variance trade-off
- Audit, test, disclose

No stage of the pipeline is bias-proof, which is exactly why catching it falls to the engineer building it.